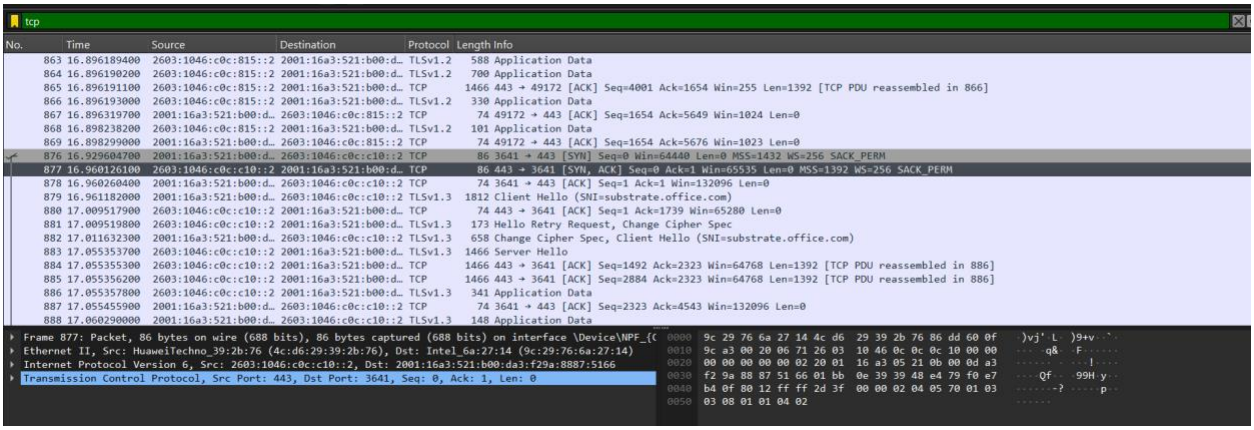
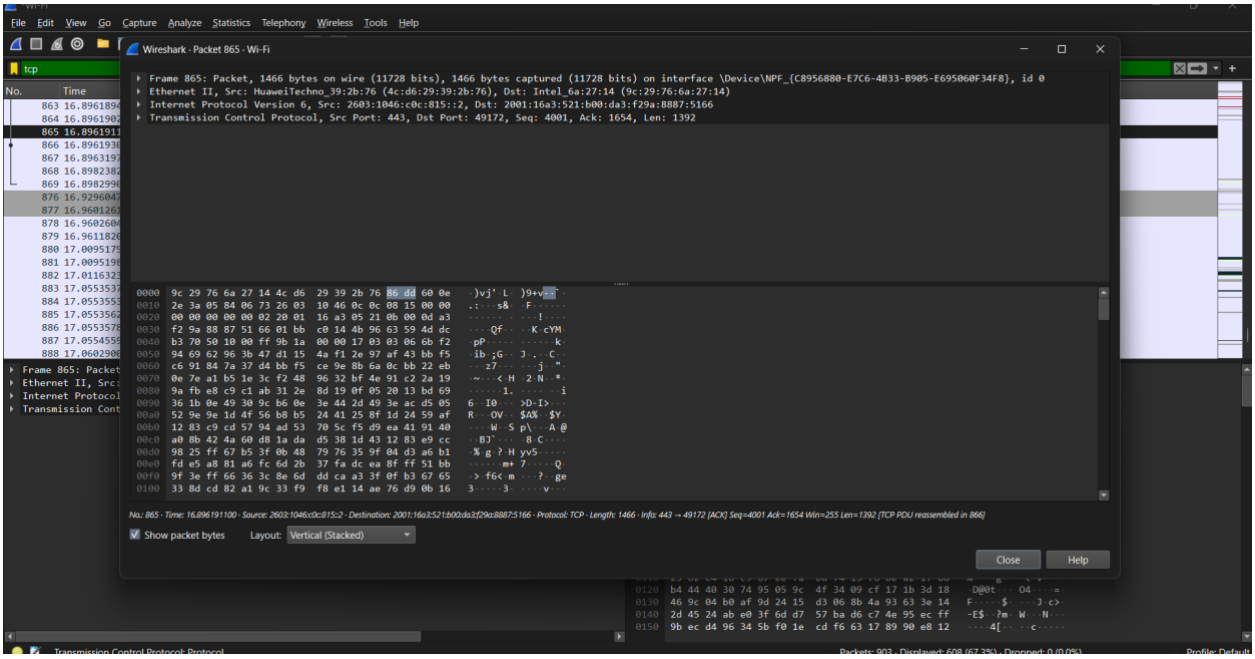
Screenshots:

1. **TCP three-way handshake**



2. **TCP data transfer**



Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented and establishes a connection using the three-way handshake (SYN, SYN-ACK, ACK). It also provides reliable delivery using acknowledgments and retransmission of lost data.
Data Integrity and Ordering	TCP	TCP uses sequence numbers and acknowledgments to ensure that data is received correctly and in the proper order. Lost or corrupted data can be retransmitted.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), email, file transfers, and other applications where reliable delivery is important.	Video/audio streaming, VoIP, online gaming, DNS, and other applications where speed and low delay are important.
Performance	Generally slower because it has connection establishment, acknowledgments, retransmissions, and other reliability mechanisms.	Generally faster and has lower overhead because it does not establish a connection or provide TCP-style acknowledgments and retransmissions.